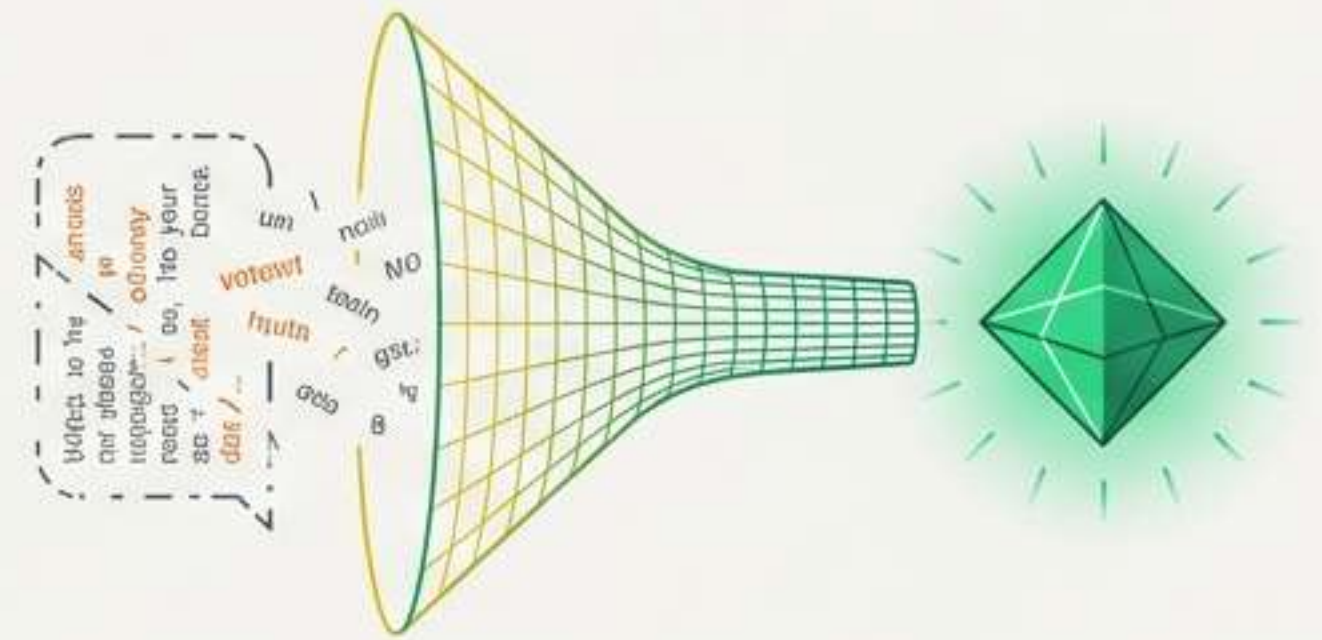


Unlocking the Financial Pulse from Mobile Money SMS Data

CSC 3221 Final Project — Introduction to Data Science



The Problem

In **Cameroon**, mobile money is the dominant retail-payment rail, but transaction data sits siloed and fragmented in users' SMS inboxes.

The Objective

Extract, clean, and model raw SMS data to predict user-activity tiers (Low, Medium, High) without compromising privacy.

The Scope

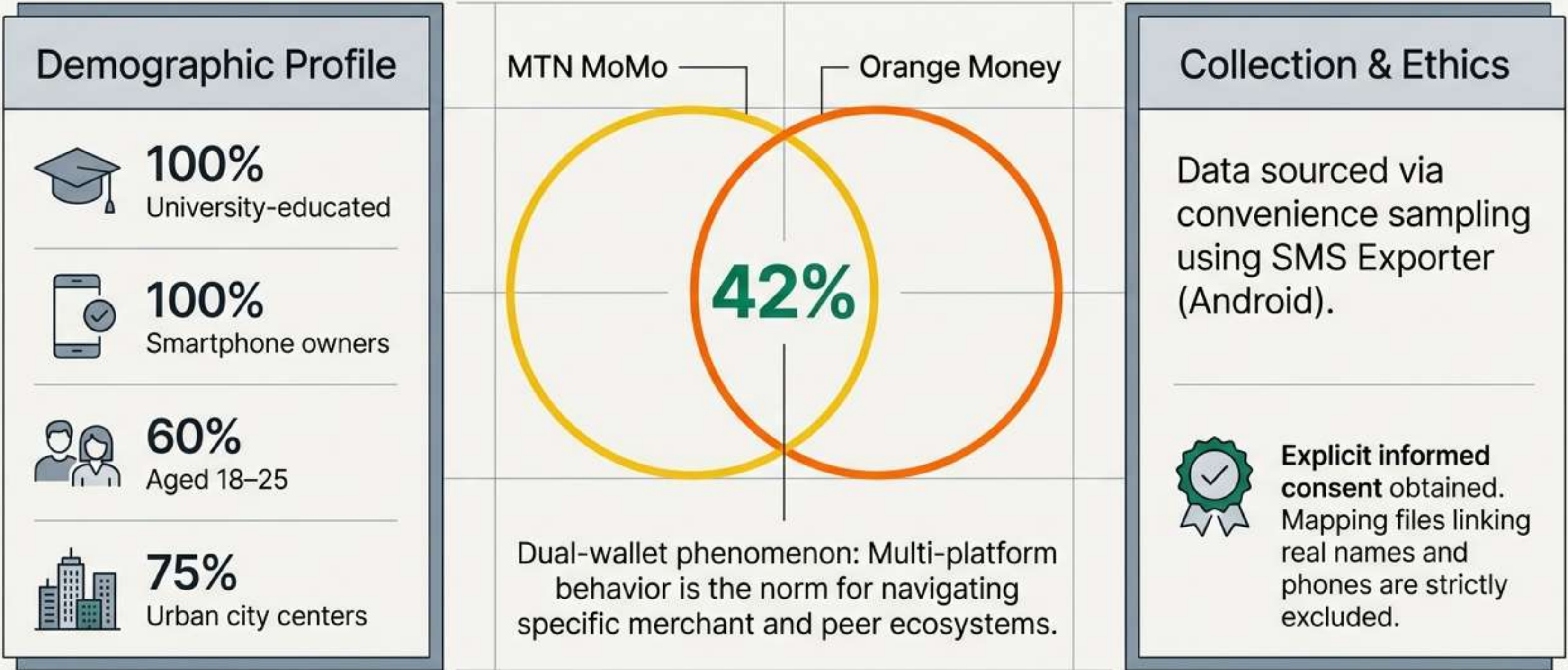
7,772

Anonymized Transactions

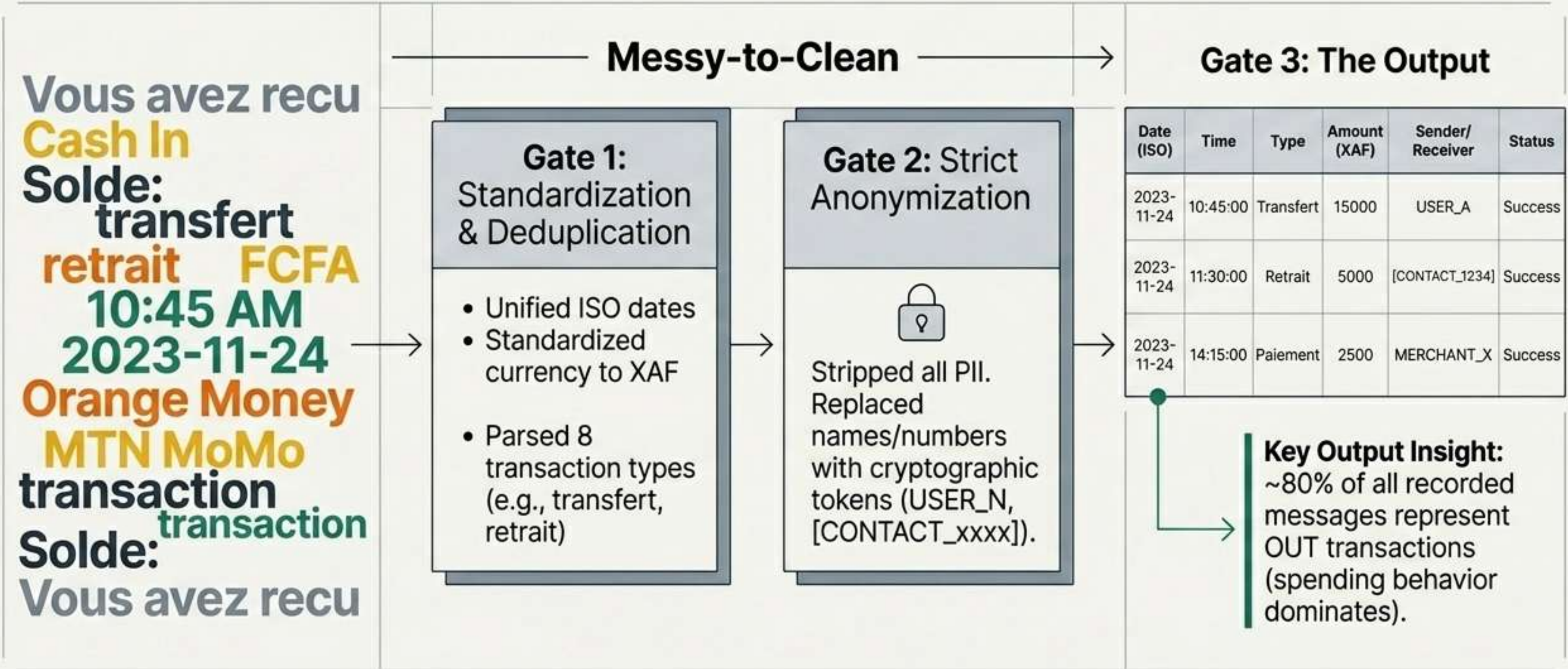
16 Consenting Users

36-Month Period

Capturing a Young, Urban, Dual-Wallet User Network



Extracting Signal from Noise via a Strict Cleaning Pipeline



Engineering Behavioral Signals from Transaction Metadata

Volume & Frequency



Features: 'tx_per_month' and 'total_amount_out'.

↳ **Impact:** Identified as the dominant signals of user engagement.

Directionality



Feature: 'send_receive_ratio'.

↳ **Impact:** Distinguishes heavy spenders from passive receivers.

Temporal Rhythms



Feature: 'weekend_ratio'.

Impact: Reveals occupational splits—salaried users skew weekday-heavy, traders skew weekend-heavy.

Financial Velocity

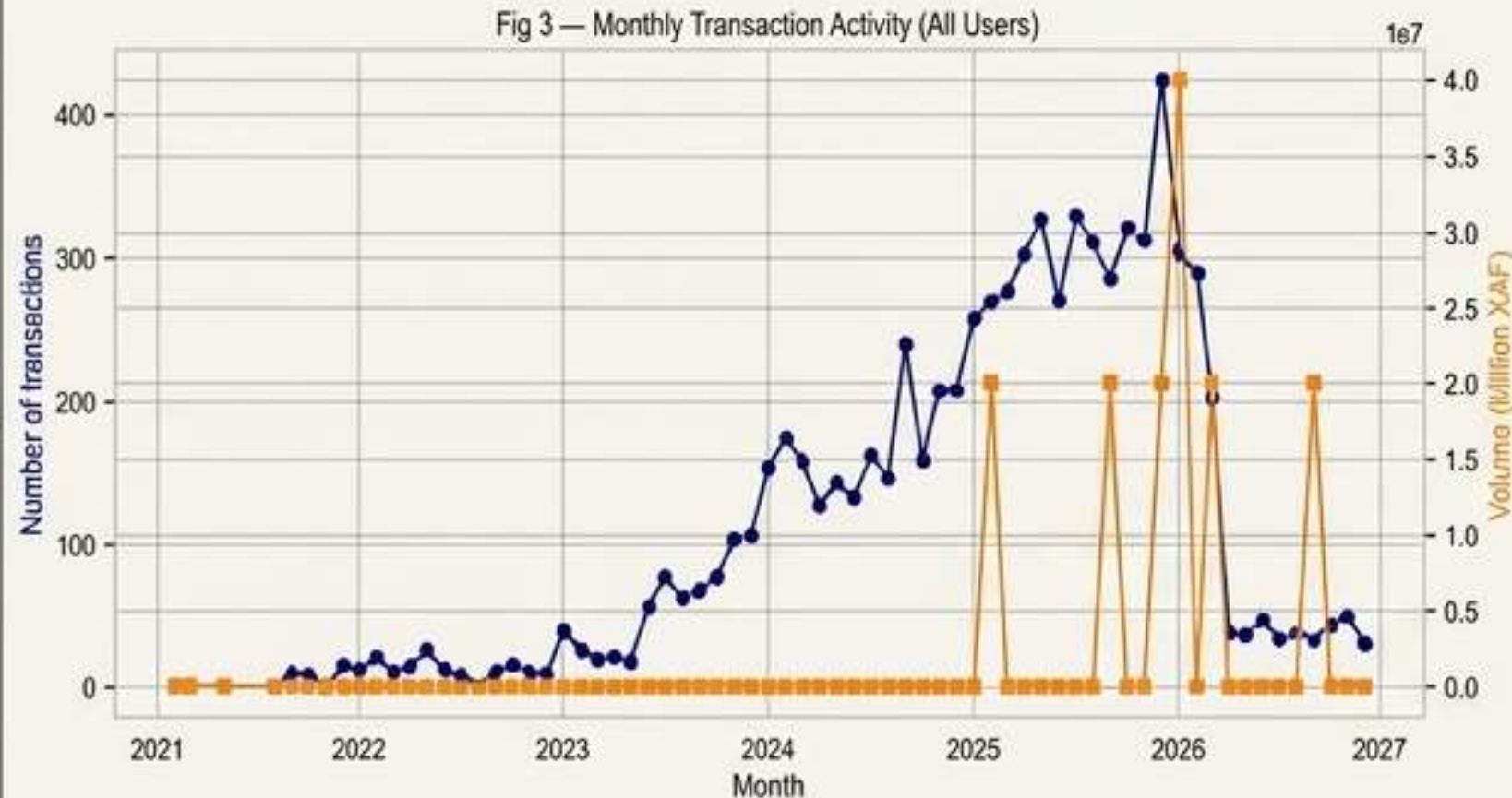


Features: 'avg_balance' and 'tx_velocity'.

↳ **Impact:** Measures the speed and retention of capital within the wallet.

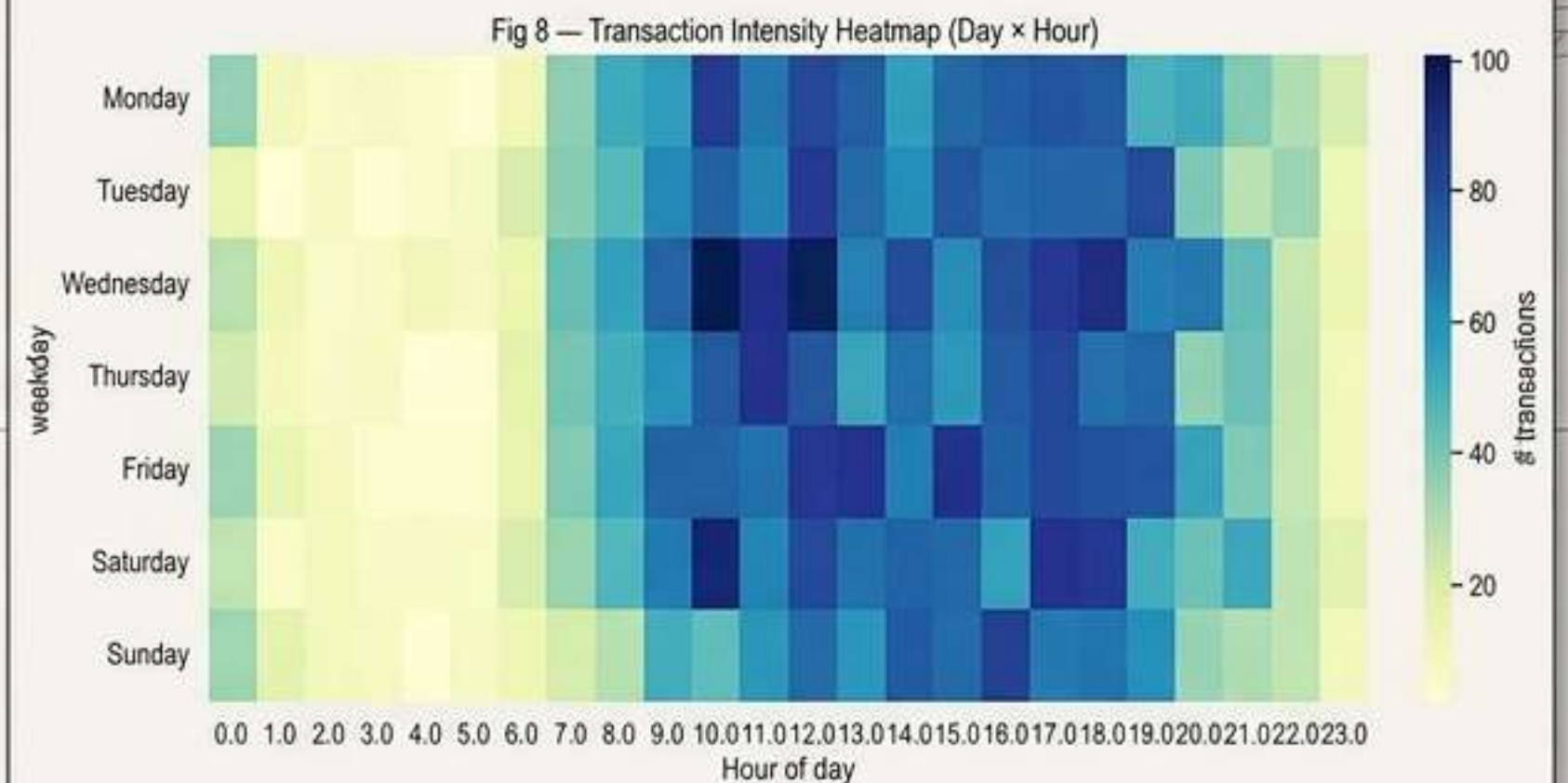
The Temporal Rhythm of Mobile Money Activity

Macro Trend: Long-Term Growth



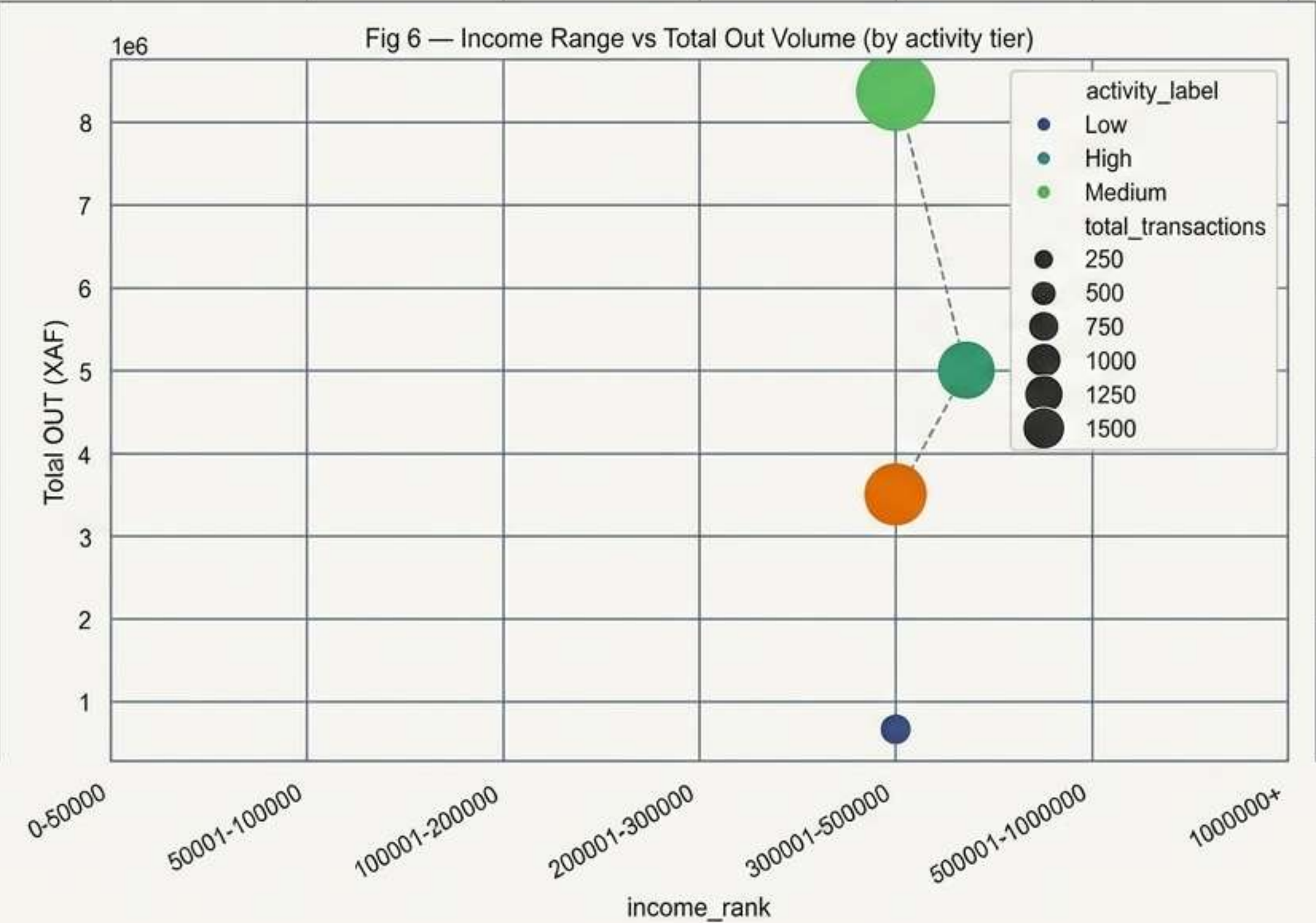
Steady, long-term growth trajectory in transaction volume, peaking in early 2026.

Micro Rhythm: Day × Hour



Key Insight: Activity is heavily concentrated during weekday business hours (08:00–19:00), with a secondary spike on Sunday afternoons. Agent float must be managed around these peak windows.

Behavioral Frequency Overrides Traditional Demographics



Insights

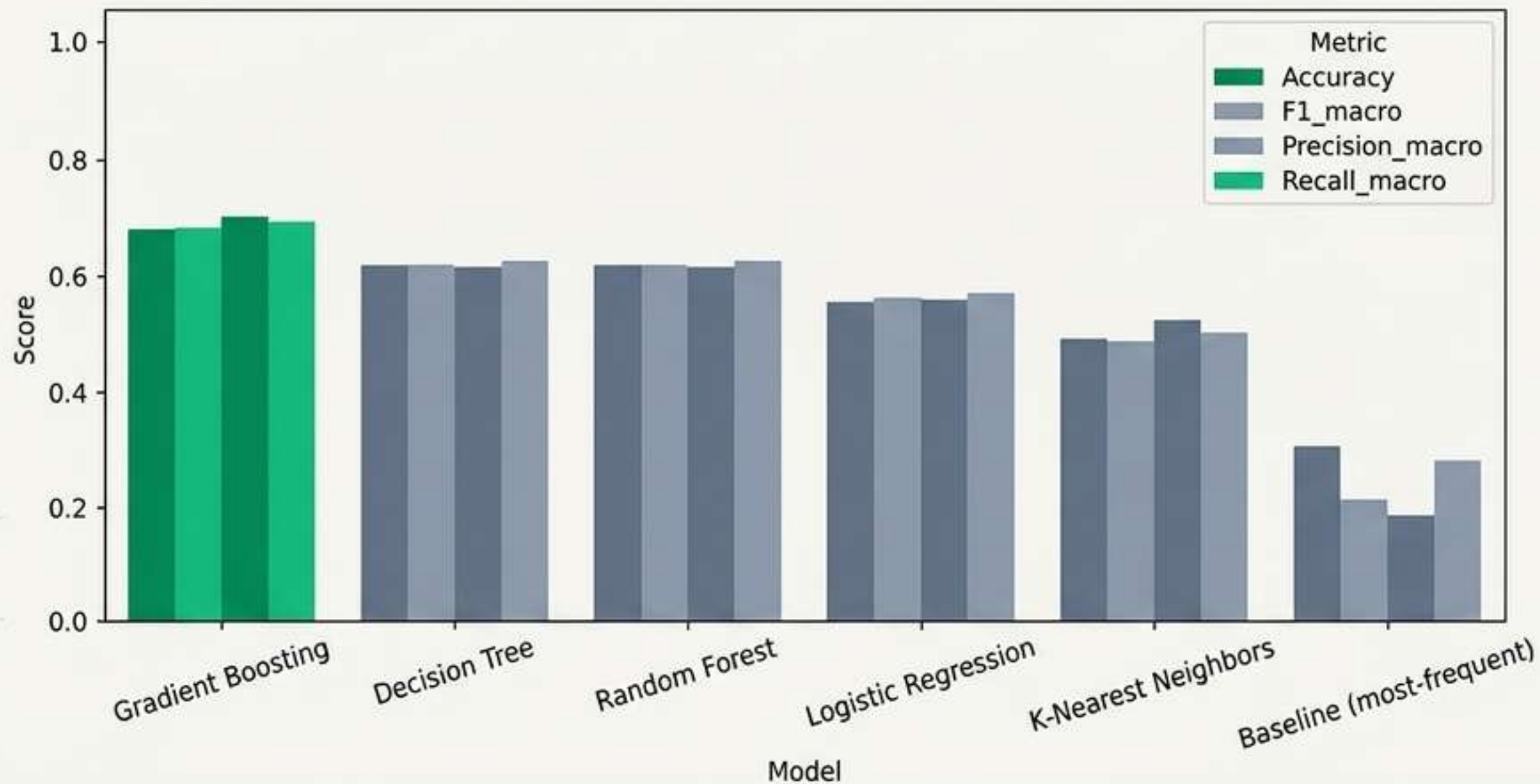
The Core Finding:

While higher income generally correlates with larger OUT volumes, the scatter plot reveals a critical non-linear relationship.

High-frequency, low-income users can match or exceed the total spend volume of mid-income users.

Conclusion: Behavioral features (transaction frequency) are vastly superior to simple demographic segmentation for predicting a user's value tier.

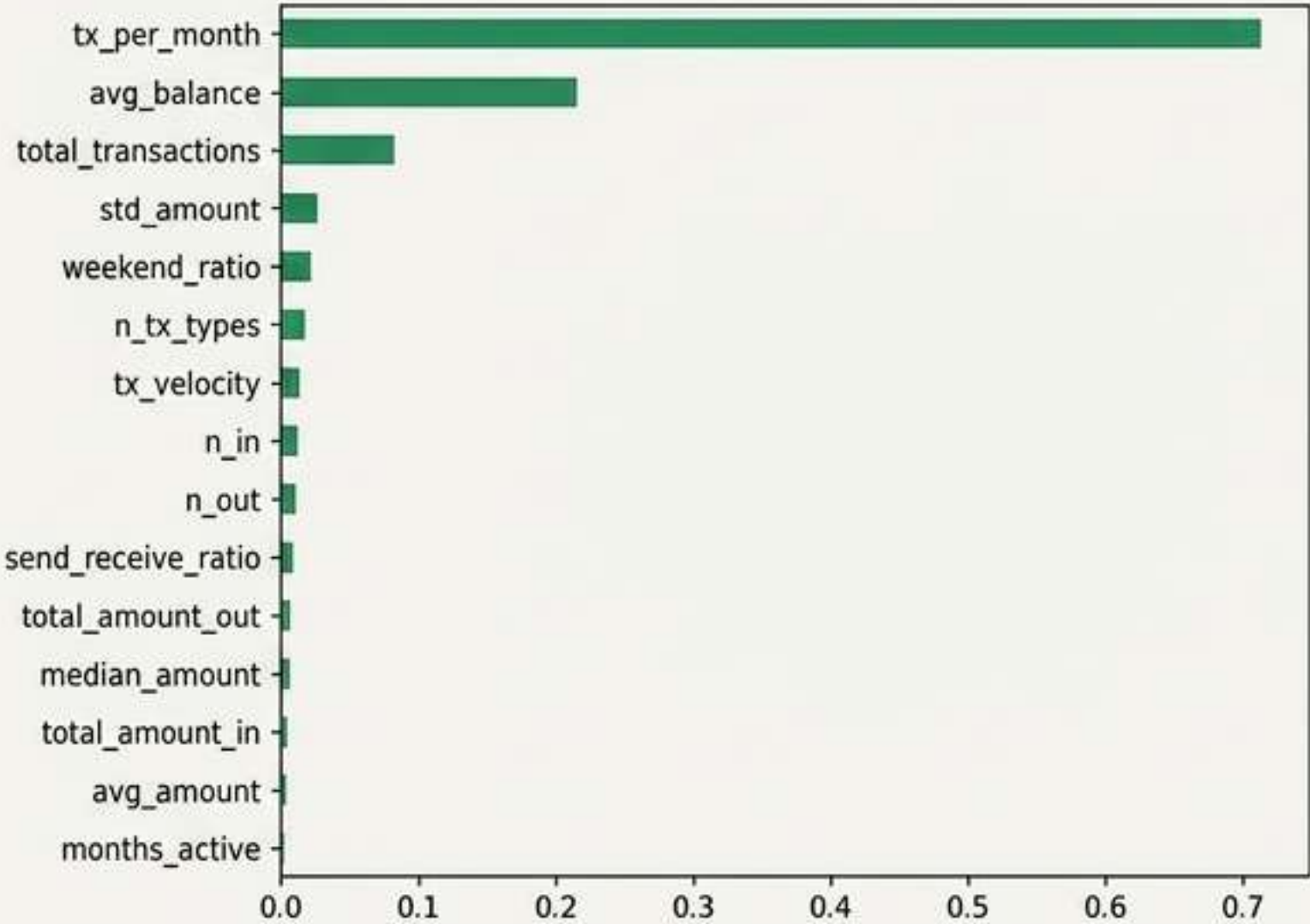
Model Evaluation: Gradient Boosting Triples the Baseline



The Contenders	The Results & Tuning
Evaluated via stratified 5-fold CV: Logistic Regression, KNN, Decision Tree, Random Forest, Gradient Boosting, and Baseline.	Baseline Macro-F1: 0.222. Gradient Boosting Macro-F1: 0.689 (Accuracy: 0.688, Precision: 0.707). Optimized via GridSearchWCV (learning_rate: 0.05, max_depth: 2, n_estimators: 100) to prevent overfitting on the n=16 sample size.

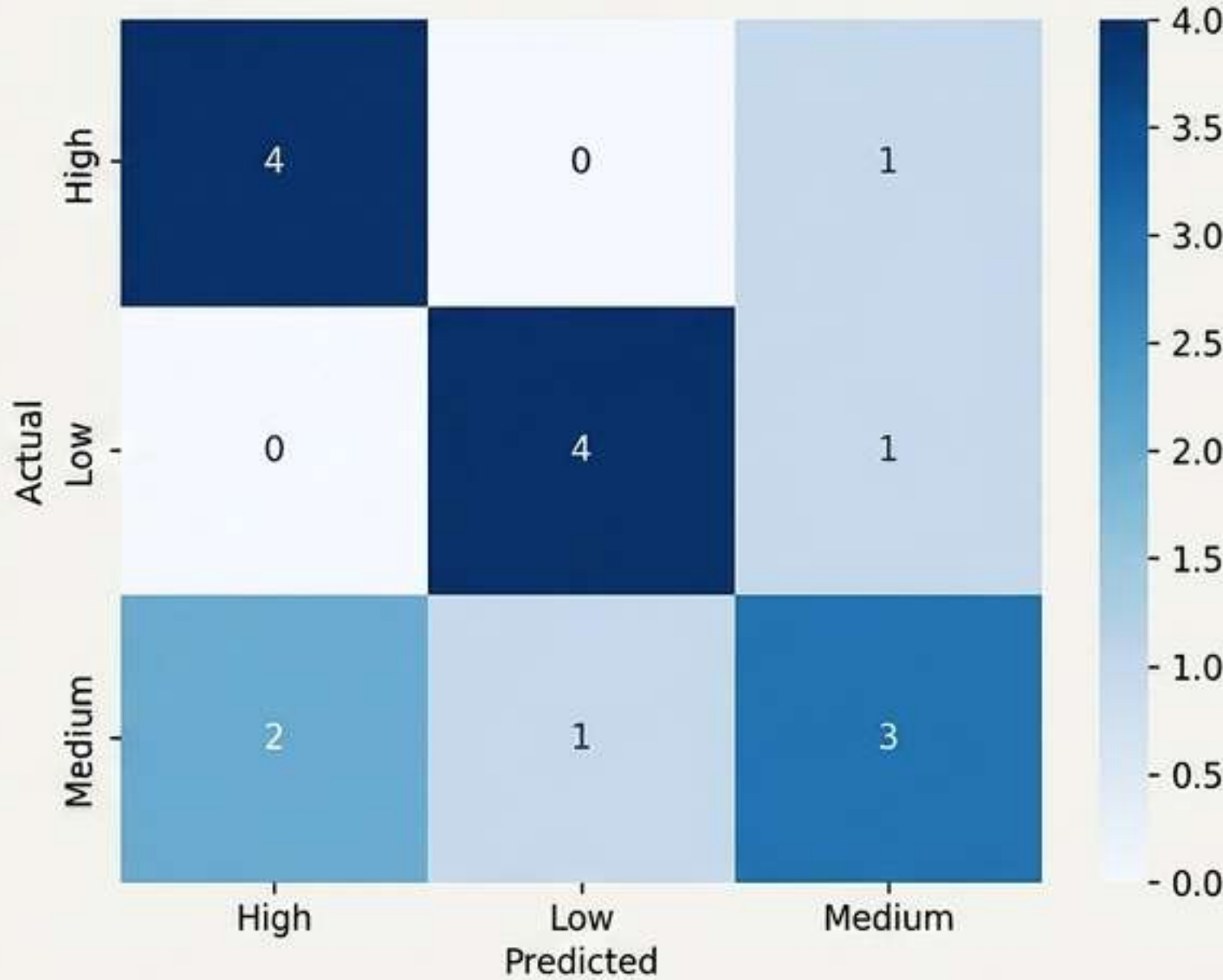
Inside the Engine: Feature Dominance and Boundary Resolution

Feature Importance



Confirms EDA hypothesis: 'tx_per_month' overwhelmingly dominates predictive power, followed by 'avg_balance'. Demographic inputs added limited marginal value.

Confusion Matrix



Shows a rigid, highly accurate boundary between Low and High users (0 confusions). Errors are exclusively clustered at the Medium/High boundary, expected given the continuous nature of the data and tercile cut-offs.

Translating Predictive Tiers into Actionable Business Strategy

Pillar 1: Targeted Product Marketing

High-tier: Deploy merchant-payment cashback to highly engaged users.

Low-tier: Deploy re-engagement bundles and guided in-app tutorials, directly addressing their low **send_receive_ratio**.

Pillar 2: Risk Pre-Screening

Use the model as a lightweight, low-cost behavioral pre-filter.

Forward only Medium/High users to expensive credit bureaus, potentially reducing query costs on rejected segments by ~35%.

Pillar 3: Operational Planning

Leverage the Day × Hour temporal rhythms discovered during EDA.



Ensure mobile money agent float is systematically replenished before the 09:00 weekday spike.

Methodological Limitations and Ethical Deployment

Guardrails & Limitations

- **Sample Size:** $n=16$ is a methodological demonstration, not a population-level claim. Features a heavy urban/educated bias.
- **Label Leakage:** Production deployment requires substituting concurrent behavior labels with forward-looking labels (e.g., next-month volume).
- **Strict Ethics:** The model must never be used for automated denial of service; human review remains mandatory for credit decisions.

“ Final Takeaway

Starting from raw, unstructured SMS exports, we built a fully privacy-respecting pipeline.

The resulting Gradient Boosting model proves that behavioral transaction rhythms are **highly** predictable, offering a **scalable engine for financial inclusion and operational efficiency.**